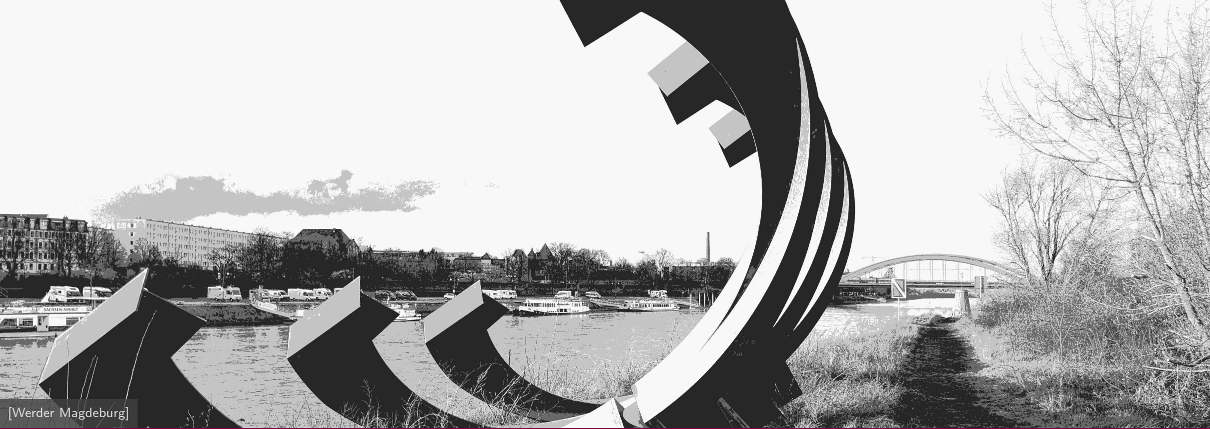

[Werder Magdeburg]

torte: Reproducible Feature-Model Experiments à la Carte

ICSE 2026 — April 12–18 — Rio de Janeiro, Brazil

Elias Kuiter[1]

University of Magdeburg[1], Germany

OTTO VON GUERICKE UNIVERSITÄT MAGDEBURG

Implementing Configurable Software Systems

A Configurable Graph

```
class Node {
  #ifdef LABELED
    std::string label;
  #endif
  #ifdef COLORED
    std::string color;
  #endif
};

class Edge {
  #ifdef DIRECTED
    Node from, to;
  #elif UNDIRECTED && HYPER
    std::set<Node> nodes;
  #endif
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```

Product Line Implementation
(here: C++ with C preprocessor)

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LABELED
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Configuration

Labeled Directed

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Colored Undirected Hyper

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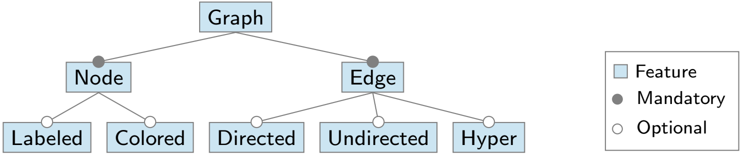
Product Implementation

Modeling Features and their Dependencies

Feature Models

- tree models **features**
- cross-tree **constraints** model dependencies
- solver-based **analyses** can be used to understand the configuration space better



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$\mathit{Hyper} \rightarrow \mathit{Undirected}$	**missing implementation**
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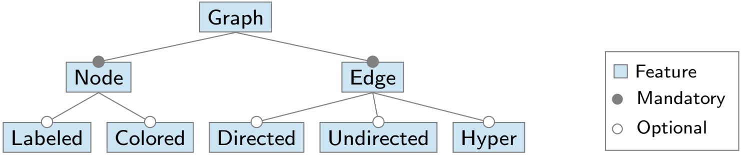

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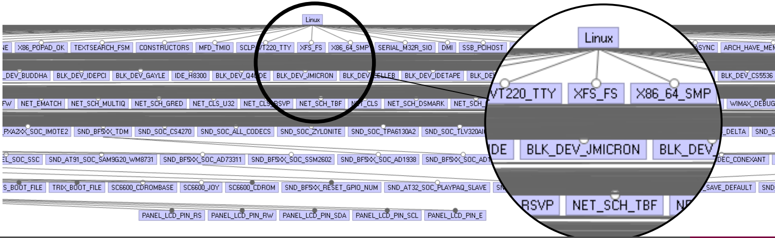
The Linux Kernel

- ≈ 20000 features [2025]
- $> 10^{1600}$ products [2025]
- 114 dead features [2013]
- 151 reverse dependency bugs [2019]



Feature-Model Analysis – Background

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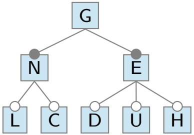
Feature Model



$$\neg(D \wedge U)$$
$$H \rightarrow U$$
$$D \not\leftrightarrow (U \wedge H)$$

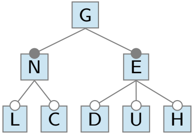
Feature-Model Analysis – Background

Model

ask

interactive

Feature Model



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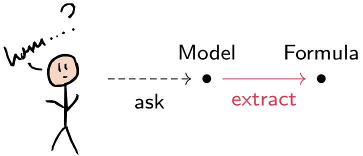
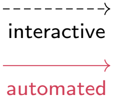
Core Feature F?

#Configurations?

Feature-Model Analysis – Background





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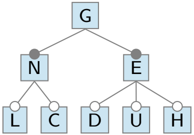

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$H \rightarrow U$
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Formula

$$
\begin{aligned}
& G \\
\wedge\ & (N \leftrightarrow G) \wedge (E \leftrightarrow G) \\
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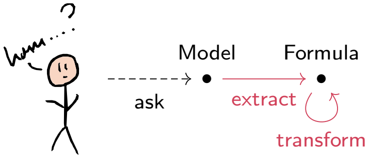

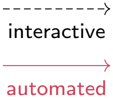

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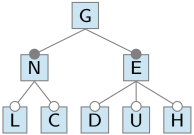

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$$

CNF

$$
\begin{aligned}
\{ & \{G\}, \{\neg N, G\}, \\
& \{N, \neg G\}, \{\neg E, G\}, \\
& \{E, \neg G\}, \{\neg L, N\}, \\
& \{\neg C, N\}, \{\neg D, E\}, \\
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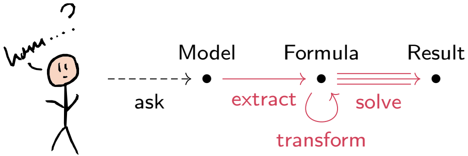

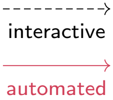

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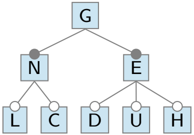

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$$SAT(\Phi \land \neg F)$$

Core Features

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#Configurations?

$$\#SAT(\Phi)$$

#Configurations

8

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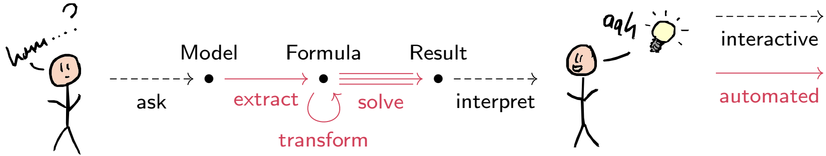

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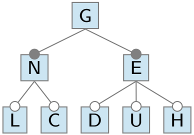

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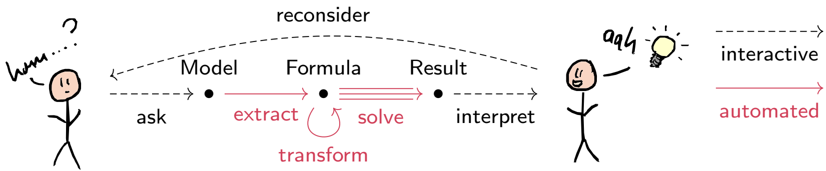

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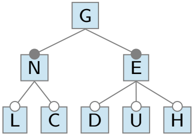

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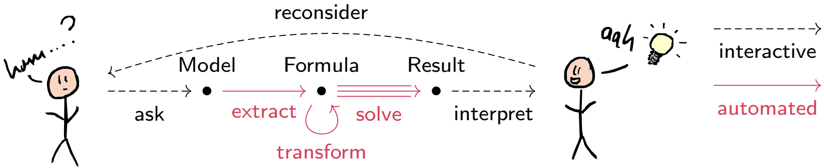

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interactive product configuration · anomaly detection and explanation · feature-model evolution and slicing · modularization · implicit constraints · code parsing · dead code analysis · code simplification · type checking · model checking · formal verification · static code analysis · dataflow analysis · testing · t-wise sampling · uniform random sampling · economical estimations · refactoring · . . .

Feature-Model Analysis – Open Challenges

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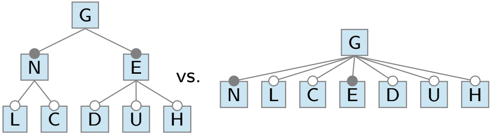

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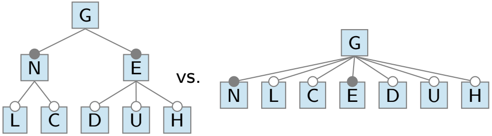

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Automation and Reproducibility

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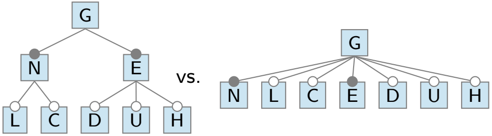

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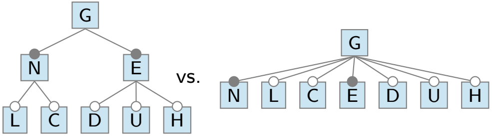

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Reusability and Interoperability

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Ease of Use

C(++) Scala Rust Java Bash Python · · ·

complex orchestration of heterogeneous tools


Introducing: **torte**!

torte – An Experimentation Platform For Feature-Model Analysis

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torte 🍰

Roll your own experiments!

/ekuiter/🍰

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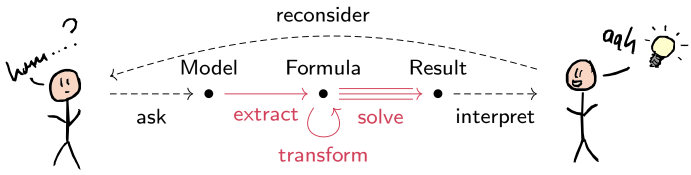

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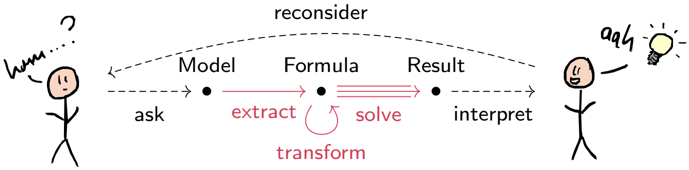

Extraction
KConfigReader KClause ConfigFix Hierarchy extractor Benchmark models

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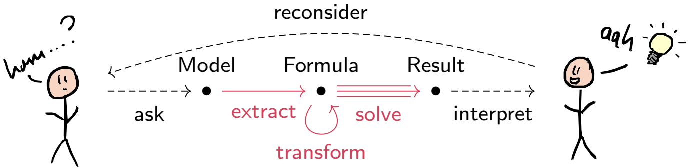

Extraction	Transformation
KConfigReader	FeatureIDE
KClause	FeatJAR
ConfigFix	clausy
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Benchmark models	CadiBack

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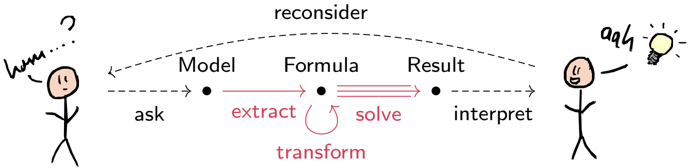

Extraction	Transformation	Solving
KConfigReader	FeatureIDE	53 SAT
KClause	FeatJAR	14 #SAT
ConfigFix	clausy	SATGraf
Hierarchy extractor	Z3	Jupyter notebooks
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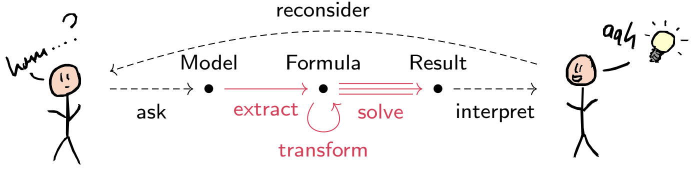

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Experimentation Platform		
Docker containerization	Job parallelization	Stage architecture
Continuous integration	Profiling system	Experiment DSL
Reproduction packages	Resource limiting	Remote execution

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Roll your own experiments!

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torte – Short Demonstration

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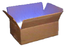
How configurable is BusyBox?

torte – Short Demonstration

How configurable is BusyBox? 

Self-Executing One-Liner

```
curl -sL https://elias-kuiter.de/torte/ | sh
```

Requirements Docker, Git, curl, and GNU tools

torte – Short Demonstration

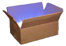
```bash
#!/bin/bash
TORTE_REVISION=7a65a22
TIMEOUT=10
JOBS=4
[[ $TOOL != torte ]] && builtin source /dev/stdin <<<"$(curl -fsSL \
https://raw.githubusercontent.com/ekuiter/torte/7a65a22/torte.sh)"

experiment-systems() {
  add-busybox-kconfig-history --from 1_36_0 --to 1_36_1
}

experiment-stages() {
  clone-systems; read-statistics; extract-kconfig-models

  # Serializes the feature model of BusyBox as UVL
  transform-model-with-featjar \
  --transformer transform-model-to-uvl-with-featureide \
  --output-extension uvl
  transform-model-to-dimacs

  # Counts the number of valid configurations of BusyBox
  solve-sharp-sat --timeout "$TIMEOUT" --jobs "$JOBS"
}
```

How configurable is BusyBox? 

Self-Executing One-Liner

```
curl -sL https://elias-kuiter.de/torte/ | sh
```

Requirements Docker, Git, curl, and GNU tools

torte – Short Demonstration

```bash
#!/bin/bash
TORTE_REVISION=7a65a22
TIMEOUT=10
JOBS=4
[[ $TOOL != torte ]] && builtin source /dev/stdin <<<"$(curl -fsSL \
https://raw.githubusercontent.com/ekuiter/torte/7a65a22/torte.sh)"

experiment-systems() {
  add-busybox-kconfig-history --from 1_36_0 --to 1_36_1
}

experiment-stages() {
  clone-systems; read-statistics; extract-kconfig-models

  # Serializes the feature model of BusyBox as UVL
  transform-model-with-featjar \
  --transformer transform-model-to-uvl-with-featureide \
  --output-extension uvl
  transform-model-to-dimacs

  # Counts the number of valid configurations of BusyBox
  solve-sharp-sat --timeout "$TIMEOUT" --jobs "$JOBS"
}
```

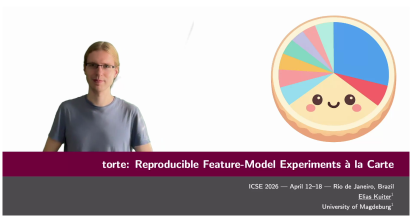
How configurable is BusyBox?

Self-Executing One-Liner

```
curl -sL https://elias-kuiter.de/torte/ | sh
```

Requirements Docker, Git, curl, and GNU tools

Demo Video



Experiences with torte

Experiences with torte

Tseitin or not Tseitin? The Impact of CNF Transformations on Feature-Model Analyses

ASE'22

How Configurable Is the Linux Kernel? Analyzing Two Decades of Feature-Model History

TOSEM'25

Investigating the Effects of T-Wise Interaction Sampling for Vulnerability Discovery in Highly-Configurable Software Systems

SPLC'25

Can SAT Solvers Keep Up With the Linux Kernel's Feature Model?

ICSE '26

Experiences with torte

- 7 Zenodo artifacts
- 3 ACM "Reusable" badges
- part of 1 best artifact award
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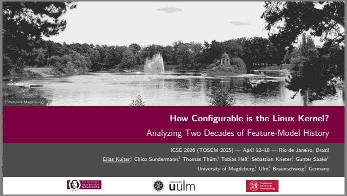
New: Review of Tool Ecosystem Beyond Torte

torte 🍰 Research

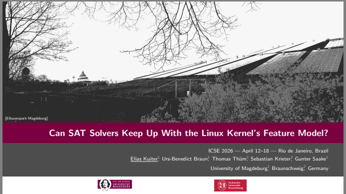
Conclusion

Journal-First Paper



How Configurable Is the Linux Kernel?

Research Paper



Can SAT Solvers Keep Up With the Linux Kernel's Feature Model?

Demo Paper

Image Credits

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